CSCC63 – Week 11 Tutorials

There are several questions here, you aren't expected to finish all of them today. But they'll be good practice for you.

1. Consider the following CLIQUE-based languages:

   (a) 100-CLIQUE $= \{\langle G\rangle \mid G$ is a graph with no cliques of size $\geqslant 100\}$.

   (b) HALF-CLIQUE $= \{\langle G\rangle \mid G$ is a graph with a clique of size $\geqslant |V|/2\}$.

   (c) CO-CLIQUE $= \{\langle G, k\rangle \mid G$ is a graph with no clique of size $\geqslant k\}$.

   (d) MAXIMUM-CLIQUE $= \{\langle G, k\rangle \mid G$ is a graph whose largest clique has size $k\}$.

   (e) MAXIMAL-CLIQUE $= \{\langle G, k\rangle \mid G$ is a graph with a maximal clique of size $k\}$.

   (f) COMMON-CLIQUE $= \{\langle G, H\rangle \mid G$ and H are graphs with maximal cliques of the same size$\}$.

   (g) 50-REG-CLIQUE $= \{\langle G, k\rangle \mid G$ is a 50-regular graph with a clique of size $\geqslant k\}$.

   (h) K-CLIQUE-DECOMPOSITION $= \{\langle G\rangle \mid G$ can be partitioned into k disjoint cliques$\}$.

   Note: *A clique is maximal if it can't be extended by adding more nodes. A clique is a maximum clique if there is no other clique of larger size.*

   A graph is d-regular if every vertex has degree d.

   Some of these problems are in $\mathcal{P}$, some are $\mathcal{NP}$-complete, some are co-$\mathcal{NP}$, and some are (probably) none of the above.

   Try to decide which ones are which, and give a short justification as to why. If you have time, write a reduction for the $\mathcal{NP}$-complete problems.

   We'll be running this as a discussion, so try to come up with your own CLIQUE variations, and discuss what their complexities might be.